



Final Exam
2024/2025

المقرر: الوسائط المتعددة
الشعبة: معتم حاسب

5/1/2024

Level: 3
Time: 2 h



Answer the following questions (90 Marks)
Exam Consist of Three Questions
Exam Questions in Two Pages

Question 1: Define The Following Terms (4 points for each one)

1. Light
2. Luminance
3. Radiance
4. Brightness
5. Depth
6. Resolution
7. Dithering
8. Image Entropy

Question 2: what is the difference between? (16 Marks)

- 1-Monochrome and Gray-scale Images (5 points)
- 2-Pseudo and True-Color Images (5 points)
- 3-Lossless and lossy compression (6 points)

Question 3: Write True or False (42 Marks)

1. Light is electromagnetic energy that radiates from a source of energy (or a source of light) in the form of waves. ☒ True
2. Visible light is in the 400 nm – 800 nm range of electromagnetic spectrum. ☒ False
3. Luminance is a measure of the light strength that is actually perceived by the human eye. ☒ True
4. Radiance is a measure of the light strength that is actually perceived by the human eye. ☒ False

5. Luminance is a measure of the total output of the source; luminance measures just the portion that is perceived. ✓ True
6. Brightness is a subjective, psychological measure of perceived intensity. ✓ False True
7. Brightness is practically possible to measure objectively. It is relative. ✓ True
8. The strength of light diminishes in inverse square proportion to its distance from its source. ✓ True
9. The smallest element on a digital image is known as a pixel. ✓ True
10. A digital image doesn't consist of a matrix of pixels. ✓ True
11. The depth of an image is the number of bits used to represent each pixel. ✓ True
12. In Gray-scale image, each pixel is stored as a single bit. X False
13. In Monochrome image, each pixel is usually stored as a byte. X False
14. In (pseudo) color image, One byte for each pixel. ✓ True
15. RGB Color Model consists of four colors. X False
16. RGB Color Model consist of Red, Green, Yellow colors. X False
17. Resolution measures how much detail an image can have. ✓ True
18. Image resolution is the number of pixels in an image.
19. Display (Monitor) resolution — refers to number of dots per inch (dpi) on a monitor. ✓
20. Vector graphics is an abstract representation of a 2-dimensional or 3-dimensional scene. ✓
21. YUV Color model is widely used in encoding color for use in television and video. ✓

مع تمنياتنا بالتوفيق والنجاح ،،،